Restricted Dog Breed Classification Using CNN Transfer Learning

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Research problem

- Benchmark six pretrained CNNs
- Binary dog-image classification
- Irish restricted-breed context
- Stanford Dogs dataset
- Controlled model comparison

Research questions

1. How do six frozen CNNs compare?
2. Does fine-tuning improve performance?
3. What does Grad-CAM reveal?
4. What limits real-world use?

Stanford Dogs Restricted-breeds

Represented	Not represented
Rottweiler	American Pit Bull Terrier
German Shepherd	English Bull Terrier
Doberman Pinscher	Japanese Akita
Bull Mastiff	Japanese Tosa
Rhodesian Ridgeback	Bandog
Staffordshire Bull Terrier	

Dataset construction

6
restricted breeds

vs.

24
unrestricted breeds

24 breeds reproducibly selected
from the remaining 114 Stanford categories

- Broader negative class
- Manageable computational scope
- Fixed data splits

Experimental pipeline

Dataset construction

Resize and normalise

Training-data augmentation

Frozen CNN benchmark

Fine-tune selected CNN

Grad-CAM explainability

Stage 1: frozen benchmark

Stage 2: fine-tuning

Stage 3: explainability

ILSVRC Winners Record

Year	Architecture	Key idea
2010	SIFT/LBP + SVM	Hand-crafted features
2011	Fisher vectors	Best pre-CNN
2012	AlexNet	Deep-CNN breakthrough
2014	VGG16	Uniform deep CNN
2015	ResNet50	Residual connections
2015	InceptionV3	Multi-scale processing
2016	Xception	Separable convolutions
2016	InceptionResNetV2	Inception plus residual
2017	NASNetMobile	Neural architecture search

Six benchmarked CNN models

- **VGG16**

  Uniform deep baseline
- **ResNet50**

  Residual connections
- **InceptionV3**

  Multi-scale processing
- **Xception**

  Separable convolutions
- **InceptionResNetV2**

  Multi-scale residual model
- **NASNetMobile**

  Architecture-search model

Frozen Benchmark Approach

- ImageNet weights
- Frozen feature extractors
- Common classification head
- Class weighting
- Early stopping
- Best validation checkpoint

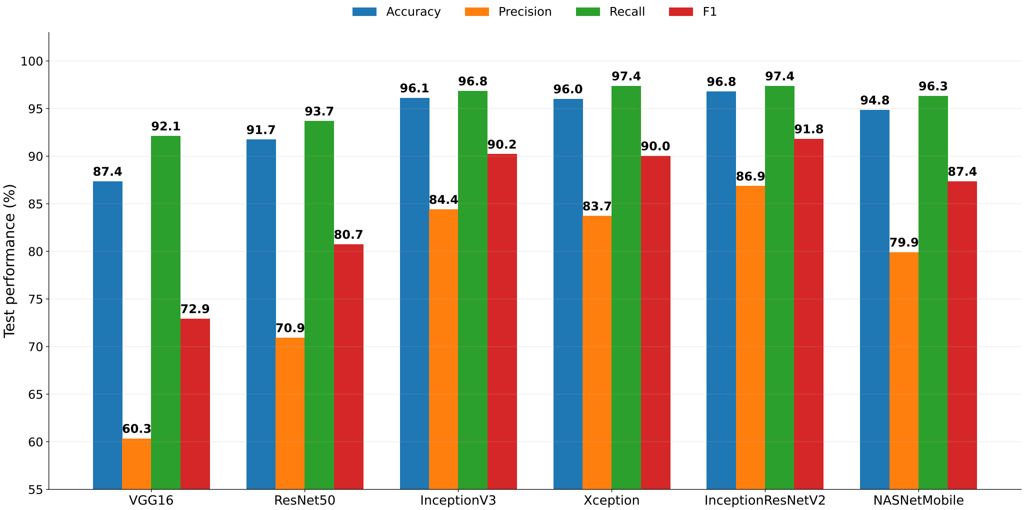
Multiple metrics—not accuracy alone

Frozen Benchmark Results Table

CNN	Acc.	Prec.	Recall	F1
VGG16	87.4	60.3	92.1	72.9
ResNet50	91.7	70.9	93.7	80.7
InceptionV3	96.1	84.4	96.8	90.2
Xception	96.0	83.7	97.4	90.0
InceptionResNetV2	**96.8**	**86.9**	**97.4**	**91.8**
NASNetMobile	94.8	79.9	96.3	87.4

Percentages shown; bold and shading are selected model.

Frozen Benchmark Results Chart



Fine-tuning strategy

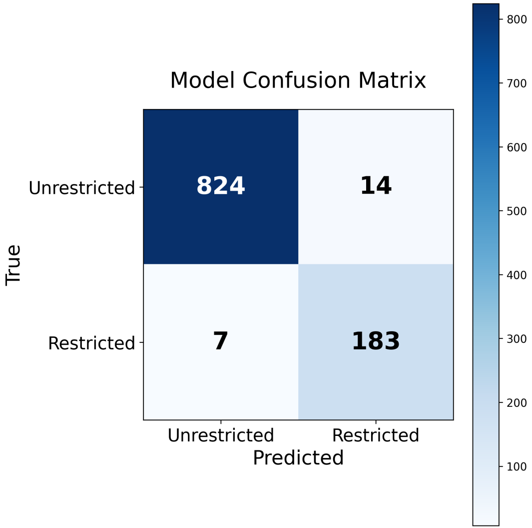
Task adaptation without destabilising pretrained features

- Last 20 layers unfrozen
- Learning rate: 10^{-5}
- Maximum 15 epochs
- Class weighting retained
- Early stopping
- Best validation checkpoint

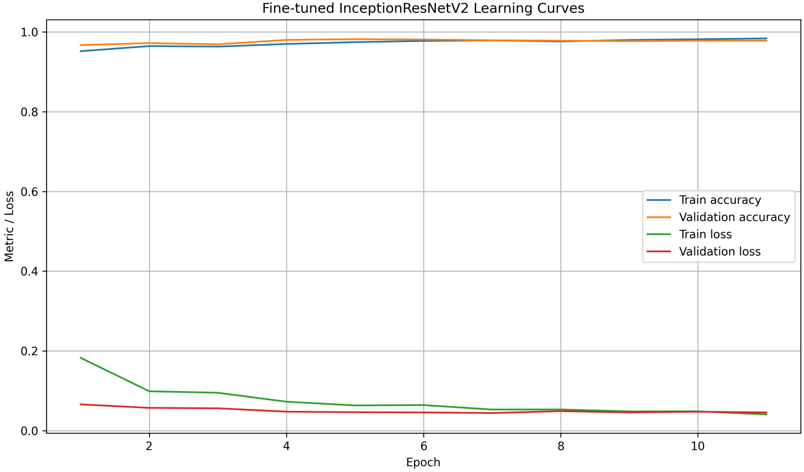
Fine-tuning results

Metric	Frozen	Fine-tuned
Accuracy	96.79%	**98.25%**
Precision	86.85%	**93.88%**
Recall	**97.37%**	96.84%
F1	91.81%	**95.34%**
MCC	0.9005	**0.9428**

Fewer false positives
Slightly more false negatives

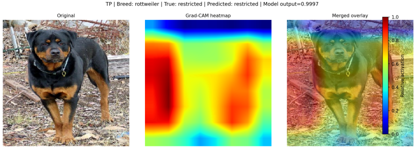


Learning behaviour

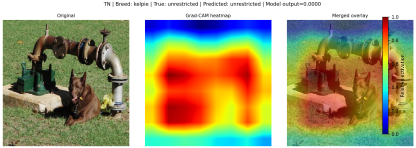


Best validation checkpoint retained

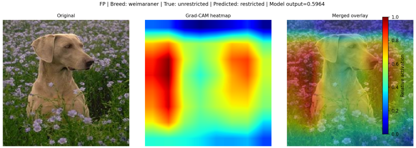
Grad-CAM explainability



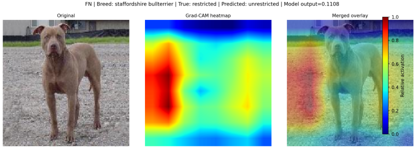
True positive



True negative



False positive



False negative

Diagnostic evidence—not proof of causal reasoning.

Project Contribution

- Reproducible pipeline
- Six-CNN benchmarks
- Multi-metric model selection
- Fine-tuning analysis
- Error trade-off analysis
- Grad-CAM explainability

IPYNB Notebook Execution

1. `01_Data_Preparation.ipynb` **CPU**
2. `02_Benchmarking.ipynb` **GPU required**
3. `03_Fine_Tuning.ipynb` **GPU required**
4. `04_Gradcam.ipynb` **CPU**

Local hardware did not provide a suitable GPU, so Google Colab was used to accelerate benchmarking and fine-tuning with a **GPU hardware accelerator** (NVIDIA T4 when available). Free-tier compute limits occasionally restricted GPU access, so some runs were completed during off-peak periods.

Limitations?

Deployment?

Further Questions?

Conclusions

1. Transfer learning performed strongly
2. InceptionResNetV2 ranked first
3. Fine-tuning improved overall balance
4. Fine-tuning changed the error trade-off
5. Current limitations do not justify real-world deployment